

FINAL YEAR PROJECT — PFE

ESPRIT · 2026

PEAXIS

An AI-Powered Hiring Operating System

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Next.js

NestJS

FastAPI

PostgreSQL

Gemini

Docker

1 INTRODUCTION

General Introduction

Academic framing of the project — context, problem, objectives, and methodology.

01 Context

AI is reshaping every industry. Recruitment — still largely manual — is one of the last domains to benefit from intelligent automation.

02 Problem Statement

Traditional ATS tools are keyword-driven and rigid. Hiring teams waste hours on manual screening while qualified candidates go unnoticed.

03 Objectives

Design and build PEAXIS — an AI-native, full-stack recruitment platform that automates scoring, surfaces explainable insights, and scales for SMEs.

04 Methodology

6-month PFE internship at Prospector. Iterative agile cycles: requirements → architecture → full-stack implementation → AI integration → deployment.

- **Internship scope:** 6-month PFE at **Prospector** — full-stack engineering and AI systems architecture, independently designed and delivered.

2 COMPANY PRESENTATION

Prospecter — Internship Host Company

An AI-powered B2B sales prospecting platform — automating outbound pipeline generation through autonomous AI SDR technology.

MISSION

Replace manual sales prospecting with a fully autonomous AI SDR — enabling B2B companies to scale outbound revenue without scaling headcount.

AI / LLM

B2B SAAS

SALES AUTOMATION

OUTBOUND GTM



AI SDR Engine

Autonomous AI sales rep — qualifies leads and orchestrates outreach sequences at scale.



Lead Intelligence

Enriches prospects with firmographics and intent signals to prioritise high-value targets.



Outbound Automation

Multi-channel sequences with LLM personalisation — zero manual prospecting.



Pipeline Analytics

Real-time conversion tracking, A/B message testing, and CRM sync.

 **Strategic context** — Prospecter's LLM pipelines, multi-tenant SaaS, Redis caching, and background workers provided the architectural foundation transferred directly into PEAXIS.

From **AI Sales Infrastructure** to **AI Hiring Infrastructure**

A 6-month final year internship at Prospector — sole engineer for the design, architecture, and implementation work. PEAXIS is a separate product developed during the internship and is not a Prospector product.

PEAXIS is an independent product developed during the internship period at **Prospector**. It is not related to Prospector's product — it is a separate AI recruitment SaaS platform conceived, designed, and built entirely as the internship project.



Full-Stack Architecture

- NestJS API — 14 feature modules
- Next.js Hire + Jobs portals
- Multi-tenant data model



AI Integration

- FastAPI AI microservice
- CV parsing + match scoring engine
- Hiring Copilot · JD Generator



Platform Engineering

- Docker multi-stage deployment
- pgvector semantic search
- Stripe billing integration

SOFTWARE ENGINEERING INTERN

AI INTEGRATION

PLATFORM ARCHITECT

SOLO CONTRIBUTOR

3 THE PROBLEM

Recruitment is **broken** at every layer

The hiring process hasn't fundamentally changed in 20 years. Time, money, and talent are wasted at every step.

42 days

Average time-to-hire
SHRM 2024

75%

Recruiters overwhelmed by CV volume
LinkedIn Talent Report

60%

Candidates ghost after poor UX
Indeed Survey 2023



Screening bottleneck

Recruiters spend 23 hours/week manually reviewing CVs. 80% of that time is wasted on unqualified applications.



Bias & inconsistency

Manual screening introduces unconscious bias and inconsistent evaluation criteria across reviewers.



Candidate friction

Long applications, zero feedback, and opaque processes cause top talent to abandon before completion.

Global: companies lose **\$14,900** per bad hire; top candidates accept offers in under 10 days.

Regional: In Tunisia & North Africa, growing SMEs rely on spreadsheets — modern ATS adoption remains critically low.

3 THE PROBLEM

Manual workflows, legacy ATS & fragmented stacks are **not enough**

None deliver the AI-native intelligence modern hiring requires.

CAPABILITY	MANUAL	LEGACY ATS	FRAGMENTED	PEAXIS
AI-powered candidate ranking	✗	✗	partial	✓
Explainable match scores	✗	✗	✗	✓
Automatic CV parsing	✗	✗	partial	✓
Candidate-facing portal	✗	basic	basic	✓
Real-time analytics pipeline	✗	✗	✗	✓
Semantic job search	✗	✗	✗	✓
Hiring copilot / AI assistant	✗	✗	✗	✓
Multi-tenant SaaS architecture	✗	partial	✗	✓

3 THE PROBLEM

The market gap: intelligent + integrated + explainable



AI black boxes

"This candidate scored 72" — but why? Recruiters cannot act on scores they cannot explain. Bias complaints follow.



Fragmented tooling

Notion for job specs. Greenhouse for pipeline. Calendly for interviews. Zero integration — context lost at every handoff.



No candidate intelligence

Existing tools track applications. None actively guide recruiters on who to advance, how to interview, or what risks to flag.

\$1.2B

HR Tech VC funding in 2024

23%

CAGR of AI Recruitment market

78%

HR leaders plan AI adoption in 2025



PEAXIS fills this exact gap

AI scores are explainable. Tools are integrated. The candidate experience is first-class. That's the OS layer missing from the market.

4 FUNCTIONAL REQUIREMENTS

Core platform requirements

FR-01 to FR-05: foundational capabilities for a production-grade SaaS recruitment system.



FR-01

Authentication & Security

- Email/password — Argon2 hashing
- OAuth2: Google + LinkedIn
- JWT access (15m) + refresh (7d)



FR-02

Multi-Tenant Architecture

- Business-scoped data isolation
- X-Business-ID header enforcement
- Unique business slug routing



FR-03

Roles & Permissions

- Plan-based feature gating
- Free / Starter / Pro / Enterprise
- JwtAuthGuard + PlanGuard



FR-04

Job Management

- Full CRUD with rich filters
- Semantic search via pgvector
- Auto embedding generation



FR-05

Application Pipeline

- 6 stages: Applied → Hired/Rejected
- Stage transition history
- AI scoring at submission time

4 FUNCTIONAL REQUIREMENTS

AI & analytics requirements

FR-06 to FR-10: the intelligence layer — PEAXIS's core differentiator above traditional ATS tools.



FR-06

AI Match Scoring

- Deterministic skill-overlap engine
- Score 0-100 with explanation
- Strengths & weaknesses breakdown



FR-07

CV Parsing

- PDF / DOCX / TXT support
- Gemini 2.5-flash extraction
- Auto pre-fills candidate profile



FR-08

Candidate Onboarding

- Anonymous apply-with-CV flow
- Profile completeness scoring
- Semantic embedding generation



FR-09

Interview Scheduling

- Interview CRUD with type flags
- Linked to application record
- Interviewer assignment & tracking



FR-10

Analytics Dashboard

- Pipeline funnel metrics
- Score distribution charts
- Time-to-hire tracking

5 NON-FUNCTIONAL REQUIREMENTS

Production-grade **quality attributes**



Horizontal scale

NFR-01 **Scalability**

Stateless API pods behind load balancer. BullMQ handles surge via Redis queue.



OWASP aligned

NFR-02 **Security**

Argon2, JWT rotation, OAuth2, X-Business-ID isolation, OWASP-hardened API.



p95 < 200ms

NFR-03 **Performance**

Redis 3-layer caching (7d/30d/24h). Sub-200ms p95. Async BullMQ for heavy ops.



99.5% SLA target

NFR-04 **Availability**

Preferred-region SDK retries. BullMQ retry logic. Docker health checks.



Full audit trail

NFR-05 **Explainability**

Every AI score exposes: numeric value, label, matched skills, missing skills, explanation.



Microservices

NFR-06 **Modularity**

NestJS feature modules. Independent AI service. Decoupled frontend apps.

6 THE SOLUTION

PEAXIS — Integrated AI Hiring Operating System

Four deeply integrated products replacing the fragmented stack every recruiter tolerates today.



API Layer

PEAXIS Core

NestJS REST API — single source of truth. Auth, billing, multi-tenant, all data owned here.



Recruiter Dashboard

PEAXIS Hire

Next.js SPA — pipeline Kanban, AI insights panel, job management, analytics.



Candidate Portal

PEAXIS Jobs

Next.js SPA — semantic job search, CV upload & parse, application tracking.



FastAPI Service

AI Brain

Stateless Python service — CV parsing, match scoring, copilot, JD generation.

- All four products share a single Prisma → PostgreSQL data layer

NEXT.JS 16

NESTJS

FASTAPI

POSTGRESQL + PGVECTOR

REDIS

BULLMQ

GEMINI

STRIPE

PEAXIS Hire — Recruiter Dashboard

NEXT.JS 16

The workspace where recruiters live. Every hiring decision powered by AI, every action one click away.



Pipeline Kanban

Drag-and-drop stage management with AI scores visible on every card



AI Insight Panel

Match score, matched/missing skills, AI explanation per candidate



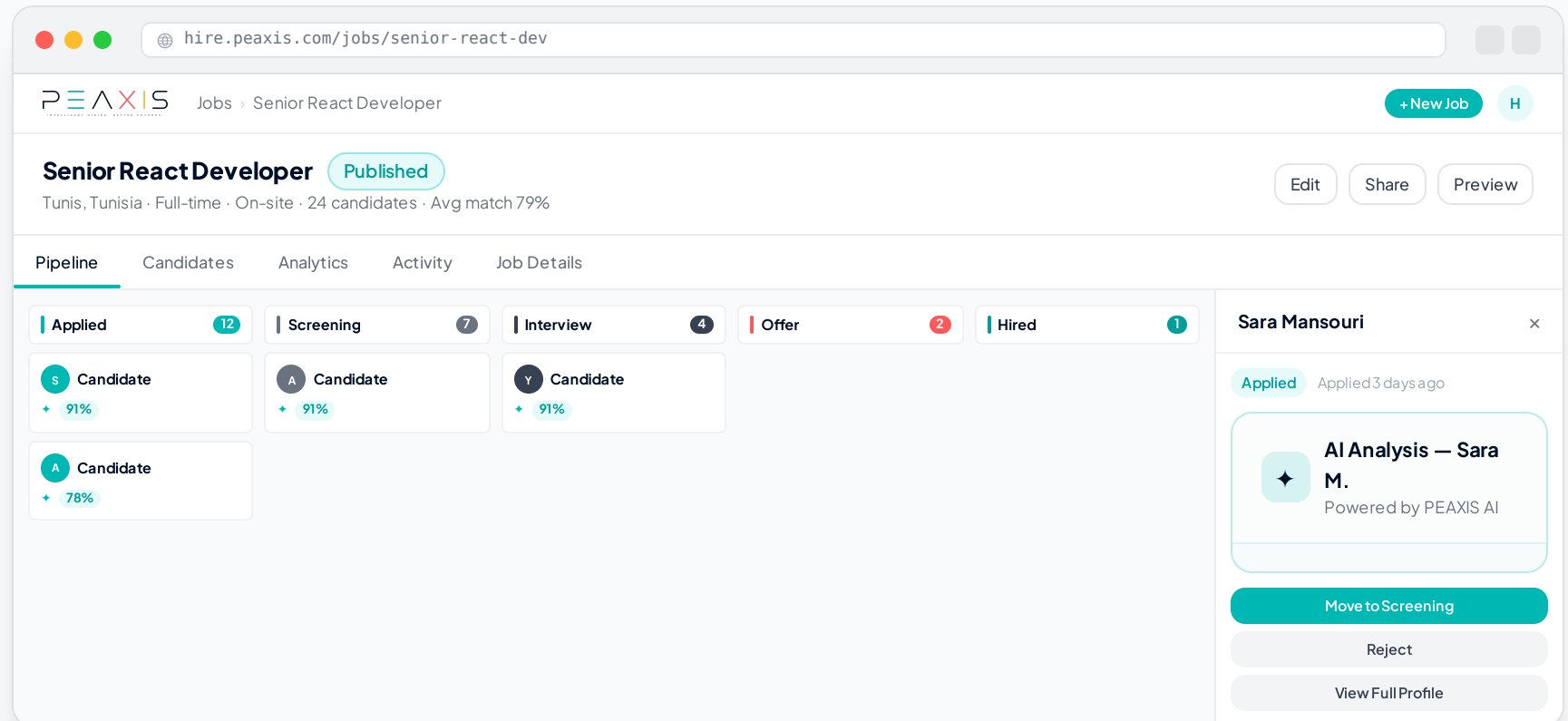
Job Analytics

Funnel metrics, score distributions, time-to-stage per job



Inline Editing

Edit job details inline without leaving the workspace — no navigation



Architecture: Server components fetch all data in parallel (Promise.all). Client components own interactive state only. Candidate card click opens a slide-in **CandidateDrawer** — never navigates away.

PEAXIS Jobs — Candidate Portal

NEXT.JS 16

The candidate experience that makes applying frictionless — from discovery to tracking.

jobs.peaxis.com

PEAXIS

Jobs Applications Profile

Sara M.

Q Search roles, companies, skills...

Search

Filters:

Tunisia

Remote OK

Full-time

AI-scored

T

Senior React Developer

TechCorp Tunisia · Tunis, TN

React TypeScript Node.js 3,500–4,500 TND

★ 91% Strong

2 days ago

Apply →

A

Python AI Engineer

AI Solutions · Remote

Python FastAPI LLM 4,000–5,500 TND

★ 84% Good

5 days ago

Apply →

D

Product Designer

DesignHub · Hybrid · Sfax

Figma UX Mobile 2,800–3,800 TND

★ 72% Good

1 week ago

Apply →

↑

Upload your CV

PEAXIS AI extracts your profile automatically

Upload CV

 **Semantic Job Search**
pgvector-powered search — finds relevant jobs even with different wording

 **Apply with CV**
Upload CV → AI parses it → profile pre-filled → one-click apply

 **Application Tracker**
Real-time pipeline stage visibility — Applied, Screening, Interview, Offer

 **Job Recommendations**
Personalised job feed powered by candidate embedding similarity

Upload CV (PDF/DOCX) → AI parses (Gemini) → Profile pre-filled → One-click apply → Real-time status

7 TECHNOLOGIES

Frontend & API stack

Frontend Layer

Landing

Marketing landing — port 3000, pnpm (node)

Rendering

Server components fetch → client components own state

PEAXIS Jobs

Candidate portal — port 3100, pnpm (node)

PEAXIS Hire

Recruiter dashboard — port 3101, pnpm (node)

Next.js

React

Tailwind

TypeScript

API Layer

Core API

14 modules — port 4000, single source of truth

Auth

JWT RS256 + Passport OAuth2 (Google, LinkedIn)

Queue

BullMQ workers for embeddings + AI scoring

NestJS

TypeScript

Prisma

JWT

Database Layer

PostgreSQL

Primary datastore — all persistent data

pgvector

1536-dim embeddings for semantic search

Redis

Cache + BullMQ queue backend

PostgreSQL

pgvector

Prisma

● Password hashing
Argon2id

● Session tokens
JWT HS256 • 15m / 7d

● Rate limiting
1000 req / 60s global

AI & infrastructure **stack**

AI FEATURE	MODEL	WHY
• CVParsing	Gemini 2.5-flash	Fast structured JSON, low cost
• Match Scoring	Deterministic	Reproducible, explainable, no LLM needed
• JD Generation	Gemini 2.5-flash	Structured output + bias checking
• Hiring Copilot	Gemini 2.5-flash	Summaries, risk flags, interview Qs
• Cover Letters	GPT-4.1-mini	Higher creative quality
• Embeddings	text-embedding-3-small	1536-dim, cost-effective

Infrastructure

FastAPI

AI service — stateless, Python 3.12+, Pydantic schemas

Redis

3-layer cache: embeddings 7d · CV parse 30d · results 24h

BullMQ

Background queue for embeddings, scoring, analysis jobs

Docker

Dev + prod compose files, multi-stage Dockerfiles

Stripe

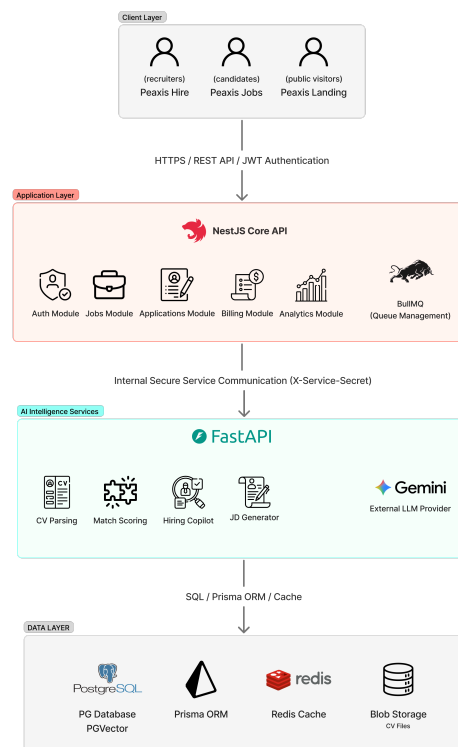
Subscription billing with webhook event handling

Redis Cache—first strategy: sha256(model+input) key → hit avoids LLM call entirely. Embeddings cached 7 days, CV parses 30 days, general AI results 24 hours.

8 ARCHITECTURE

Logical Architecture

4-layer design · Client → Application → AI Intelligence → Data

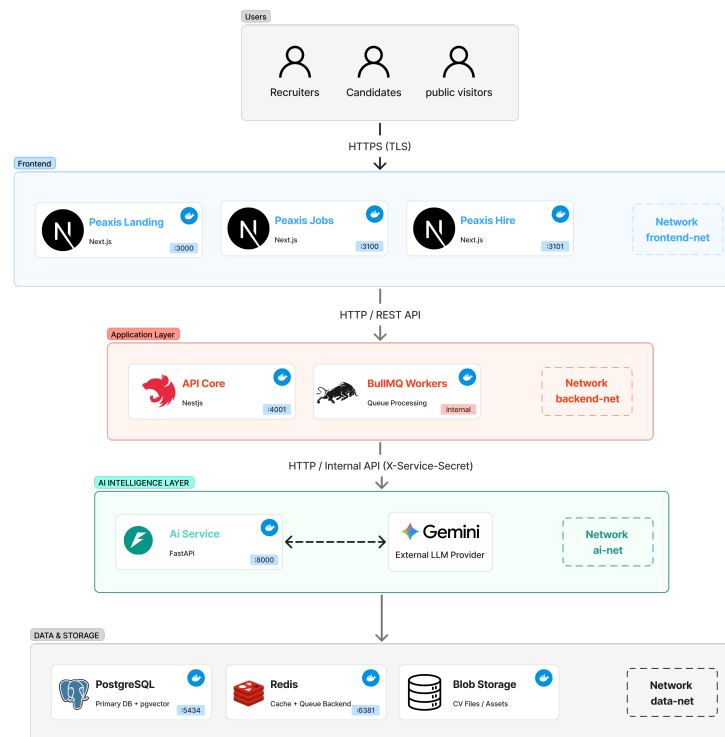


- Single Source of Truth — NestJS API owns all persistent state
- AI-Native Design — stateless AI layer returns results, never writes directly
- Modular SaaS Architecture — each layer has a single, clear responsibility

9 ARCHITECTURE

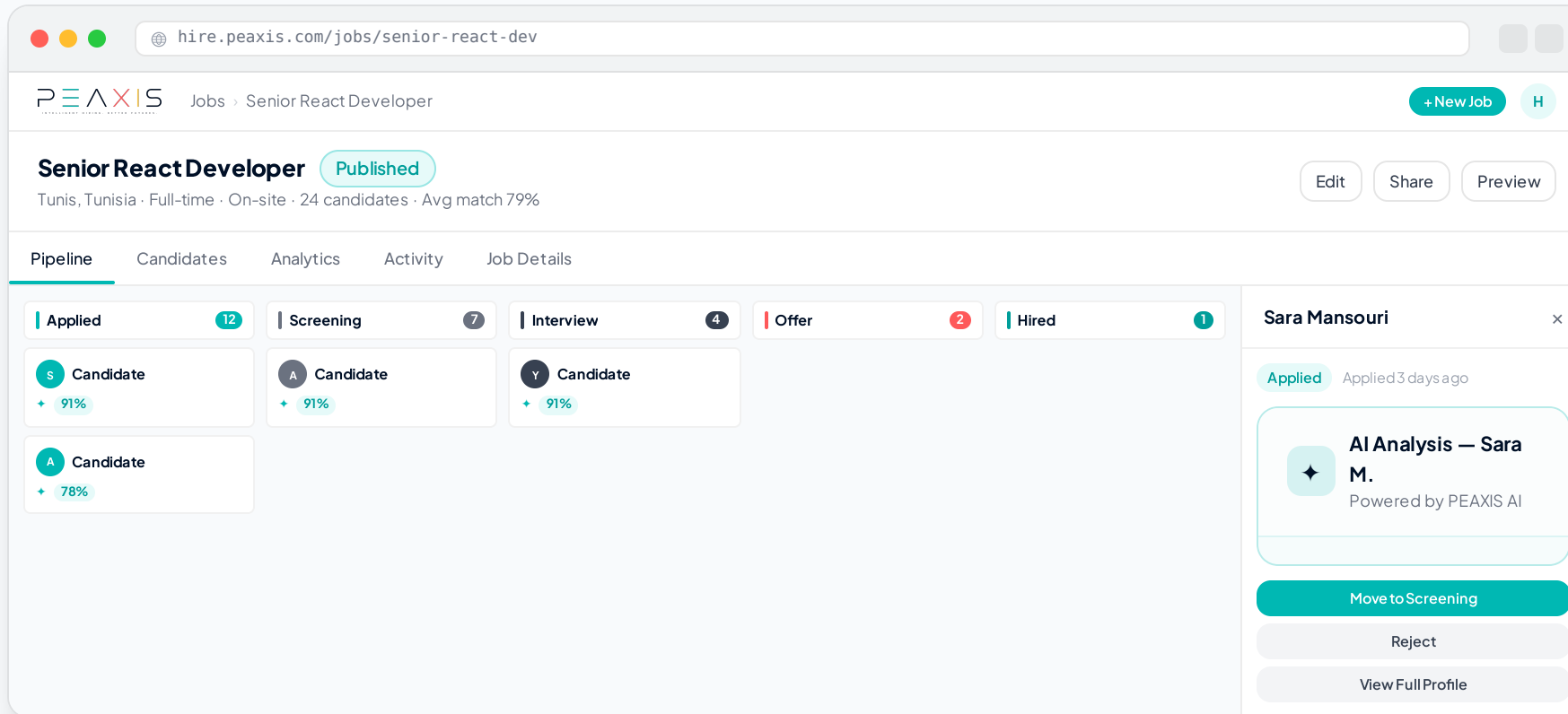
Physical Architecture

Docker Compose deployment · containerised services · network segmentation



- Containerized Services — each service runs in isolated Docker containers
- Network Isolation — frontend-net / backend-net / data-net segmentation
- Scalable Infrastructure — stateless services scale horizontally behind a reverse proxy

Demo: Recruiter Dashboard



Job Workspace

All job-related data in one tabbed view — no page hops

Pipeline Kanban

Drag-and-drop stage management with AI scores on every card

Candidate Drawer

Slide-in panel reveals AI insights without leaving the pipeline

Inline Job Edit

Edit job details within the workspace — `router.refresh()` on save

UX principle: Candidate card click *never navigates* — it opens the CandidateDrawer (slide-in panel). "View Full Profile" inside the drawer navigates. One action, clear intent.

Demo: Candidate Flow



jobs.peaxis.com

PEAXIS

Jobs Applications Profile

Sara M.

Search roles, companies, skills...

Search

Filters: Tunisia Remote OK Full-time AI-scored

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TechCorp Tunisia · Tunis, TN

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2 days ago

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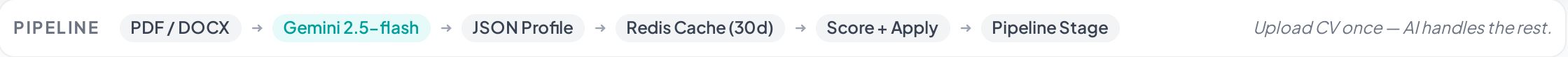
1 week ago

Apply →

Upload your CV

PEAXIS AI extracts your profile automatically

Upload CV



Demo: AI Intelligence Layer

● Match Scoring

Deterministic skill-overlap → 0–100 with matched/missing skills + plain-language explanation

● CV Parsing

Gemini 2.5–flash → structured JSON profile from PDF/DOCX in under 3 seconds

● Hiring Copilot

Candidate summary · interview questions · risk flags · next action — all in-context

● JD Generator

Bias-checked job description from title + bullets, with skill requirement suggestions

★ AI Decision Engine

INPUT

CV (PDF) + Job Description

ENGINE

Parse CV → Embed Skills

→ pgvector Match → Score + Explain

OUTPUT

82 /100

Strong Match

React ✓ TypeScript ✓ Node.js ✓

PostgreSQL ✓ GraphQL ✗ Docker ✗

"Strong React + TS background. GraphQL gap is trainable given existing Node.js depth."

Design rule: Every AI score exposes — numeric value · confidence label · matched skills (teal) · missing skills (coral) · plain-language explanation. No silent black boxes.

Technical achievements & outcomes

14

NestJS Modules

auth · jobs · apps · AI ·
billing...

60+

REST Endpoints

Swagger documented

6

AI Features

parse · score · copilot · JD

4

Microservices

Hire · Jobs · API · AI

3

Redis Cache Layers

7d · 30d · 24h TTL

100%

Multi-tenant

scoped to businessId

NOTEWORTHY IMPLEMENTATIONS

pgvector semantic search

OAuth2: Google + LinkedIn

Stripe webhook billing

BullMQ background processing

Argon2 password hashing

Docker multi-stage builds

TypeScript

65%

Python

20%

SQL/Prisma

10%

CSS/Tailwind

5%

OPERATIONAL OUTCOMES

• **Reduced screening friction**

Automated CV parsing & AI ranking
eliminates manual shortlisting

• **Faster candidate processing**

Async BullMQ workers process scoring
without blocking user actions

• **Explainable AI decisions**

Every match score includes matched skills,
gaps & rationale

• **Improved candidate UX**

Semantic search, real-time status tracking
& transparent pipeline

Challenges & lessons learned

01

CHALLENGE

Multi-tenant data isolation at scale

Every query, guard, and background job must be scoped to a businessId without leaking data between tenants.

RESOLUTION

X-Business-ID header enforced at controller level.
PrismaService always receives businessId from the guard.
BullMQ jobs include businessId in payload.

KEY LESSON

Design security boundaries first — retrofitting scoping is 10x harder.

02

CHALLENGE

AI explainability vs. LLM cost

Deterministic scoring is cheap but limited. LLM scoring is rich but expensive and non-reproducible at scale.

RESOLUTION

Hybrid approach: deterministic skill-overlap for the score (reproducible, zero cost), Gemini Flash for the explanation text (low cost, high quality, cached 24h).

KEY LESSON

Separate "what" (deterministic) from "why" (LLM). Cache aggressively.

03

CHALLENGE

Frontend state vs. server freshness

Next.js server components cache aggressively. After inline edits (job form, stage moves), UI must reflect the updated state without full navigation.

RESOLUTION

Server actions call router.refresh() after mutations — triggers server component re-fetch without navigation. Client components own transient state only.

KEY LESSON

Understand Next.js cache invalidation deeply before building complex workspaces.

12 CONCLUSION

General Conclusion

Achievements · Technical Contributions · Future Perspectives

● ACHIEVEMENTS

- 14 API modules · 60+ REST endpoints · full Swagger documentation
- 6 AI features: CV parsing, match scoring, JD generation, hiring copilot, semantic search, skill normalisation
- End-to-end multi-tenant SaaS — PEAXIS Hire + PEAXIS Jobs — production-ready
- 4 containerised microservices deployed via Docker Compose with network isolation

● TECHNICAL CONTRIBUTIONS

- AI-native queue-based pipeline: BullMQ workers → FastAPI → pgvector embeddings
- Explainable match scoring with matched/missing skill breakdowns and confidence scores
- Cache-first strategy (Redis 3-layer TTL) for AI result performance optimisation
- Secure inter-service communication via X-Service-Secret header validation

● FUTURE PERSPECTIVES

- Real-time collaborative hiring workspace with role-based decision tracking
- Multilingual CV parsing and region-aware job matching (MENA market focus)
- Advanced analytics dashboard with predictive pipeline velocity metrics
- Integration ecosystem: LinkedIn, Greenhouse, Workday — open API layer



PEAXIS demonstrates that AI-powered recruitment can be built with engineering rigour, academic depth, and real product maturity — transforming a 6-month internship into a fully deployable, multi-tenant SaaS platform ready for market.

Thank you For your time

Contact: BIBANI Mohamed Habib Allah — contact@peaxis.com · Questions welcome
after the session.

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